

SCRIBE: Learning Reusable Skill Rubrics for Process Reward Models

Yuxuan Jiang Francis Ferraro
University of Maryland, Baltimore County



yuxuanj1@umbc.edu



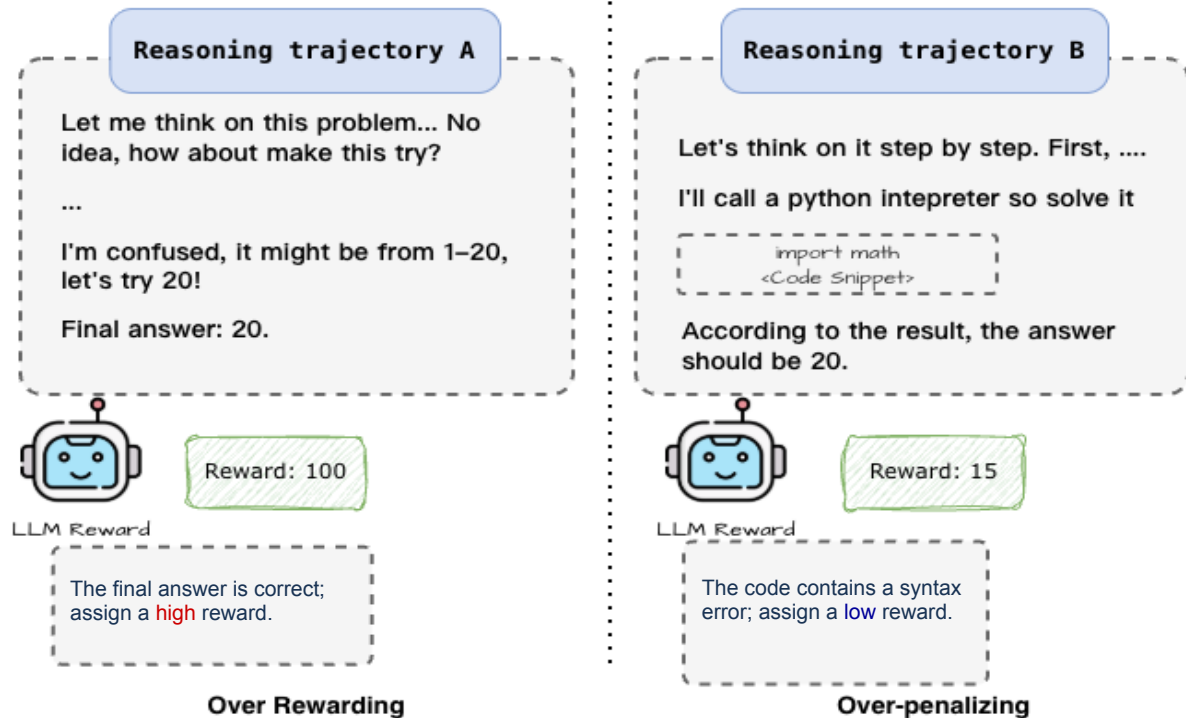
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Supported
by:



Unreliable LLM-as-a-Judge Reward



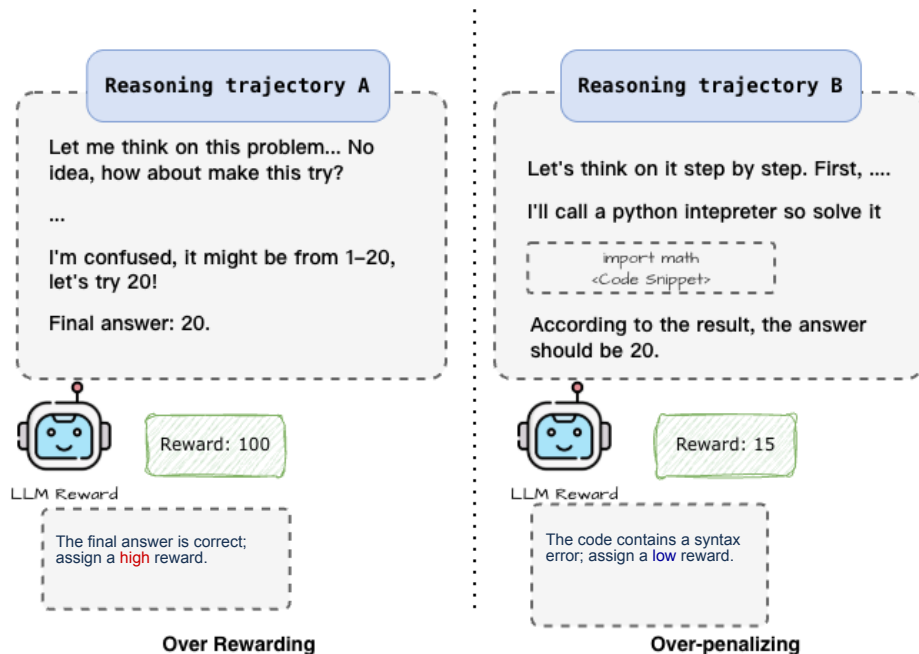
Over-rewarding:
Correct Answer but Flawed Logic

Over-penalizing:
Sound Strategy + Minor Syntax Error

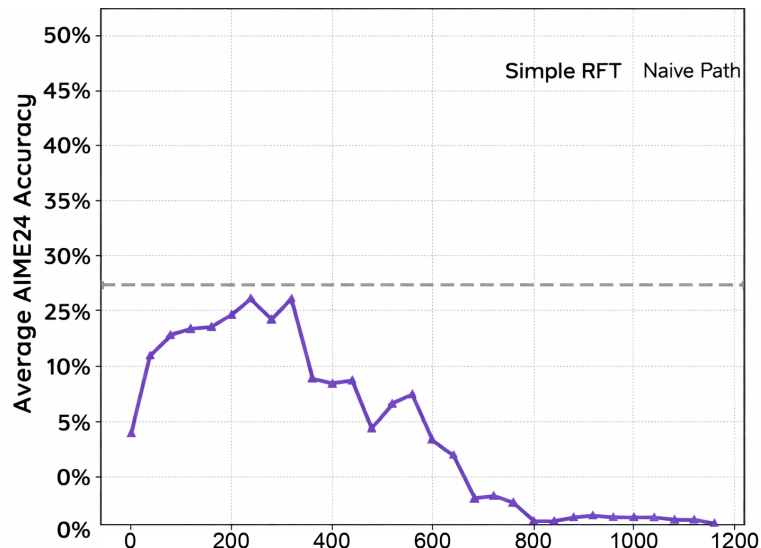
Motivation: Unreliable Reward Leads to Unreliable RL Training

Over-rewarding: Correct Answer + Flawed Logic

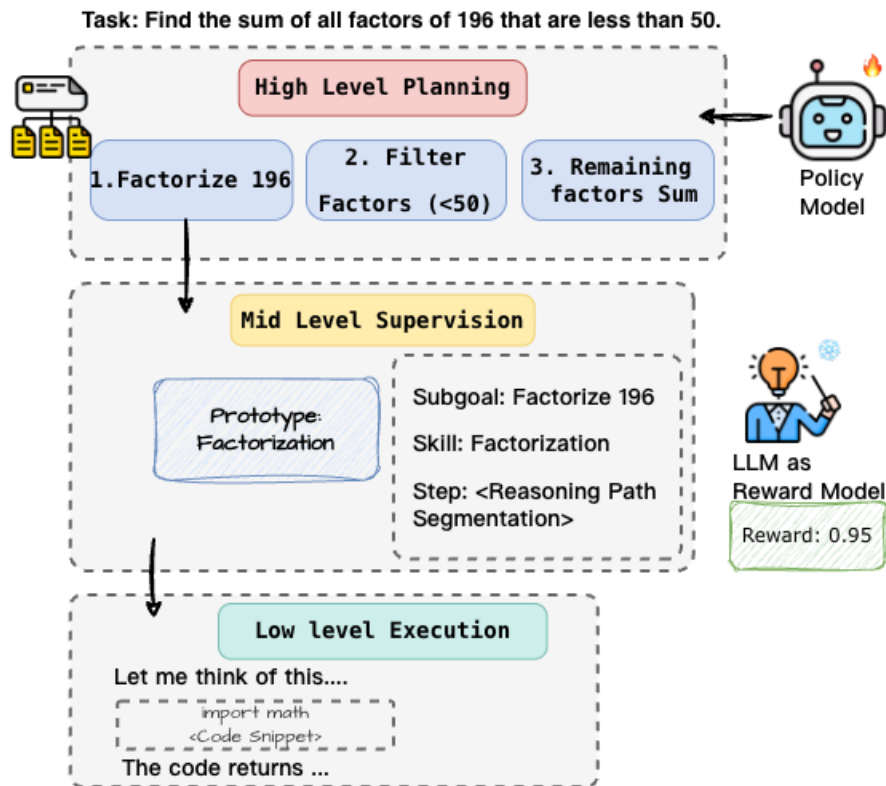
Over-penalizing: Sound Strategy + Minor Syntax Error



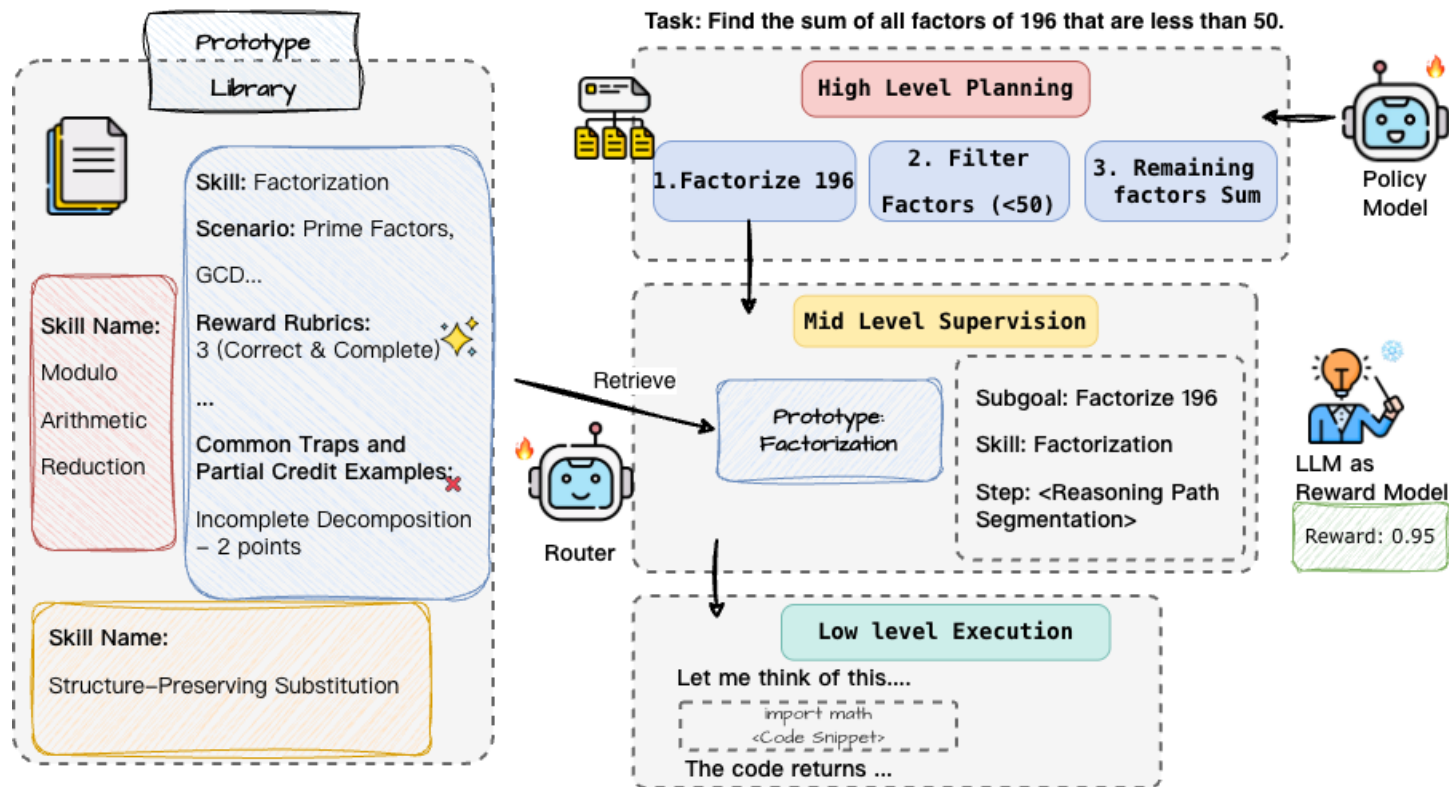
Training Crash!



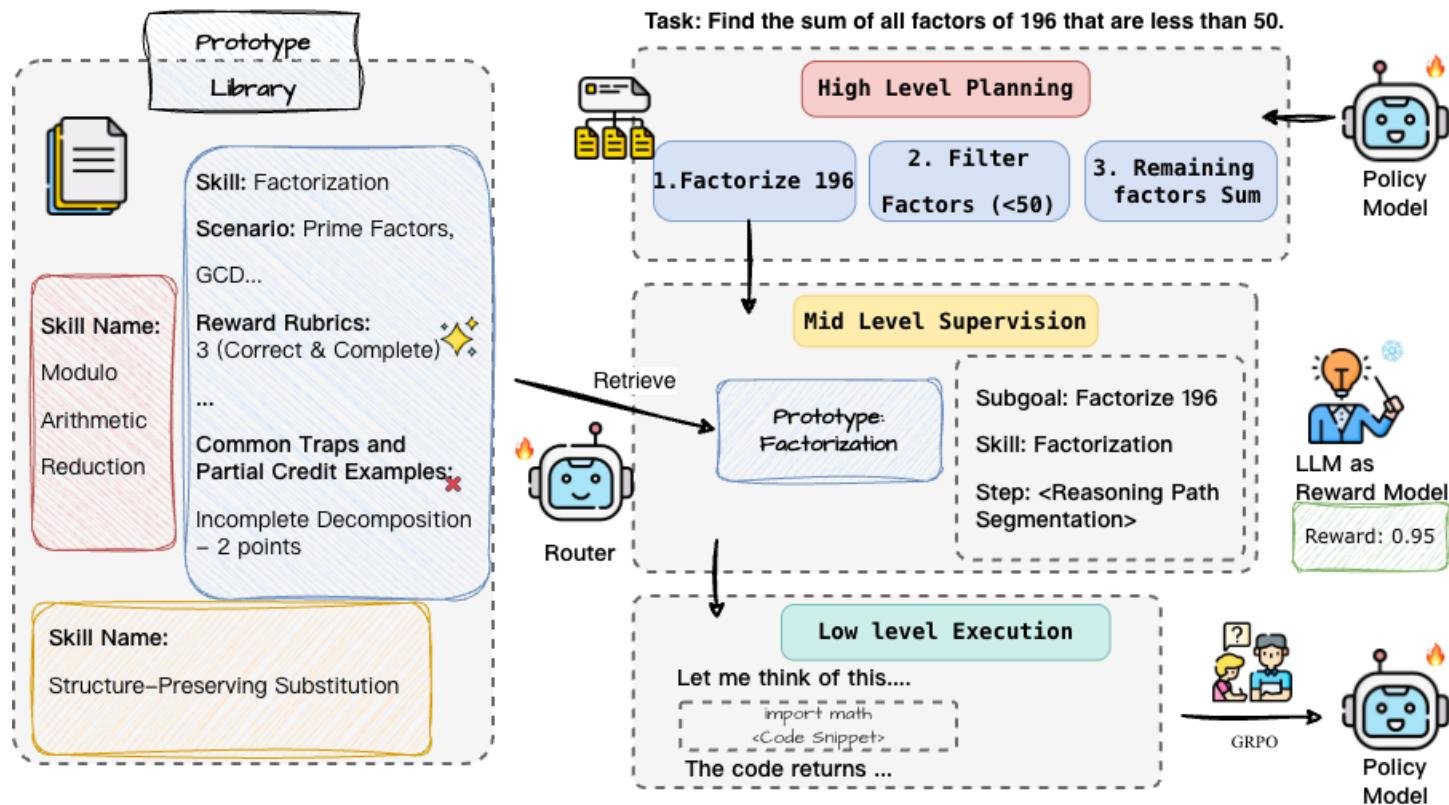
SCRIBE: Process Reward Is Not About 'Correct' but 'Done'



Skill Prototype: A Reliable Rewarding Rubric



Reliable Training



A Skill Prototype Becomes a Reusable Reward Standard

SKILL PROTOTYPE · MATHEMATICAL REASONING

Bound-Based Conclusion and Synthesis

SCOPE & APPLICABLE SCENARIO

Bounds, extrema, tightness, case coverage, and domain constraints. Conclude from prior constraints; justify attainability or feasibility.

CANONICAL REASONING PATTERN

1. Identify the relevant intermediate results.
2. Distinguish implication from a tightness claim.
3. Prove attainability when claiming an extremum.
4. Check domains and cover all cases.
5. State the final conclusion.

JUDGING RUBRIC · 0–3

3 Correct & complete: rigorous and feasible.

2 Minor flaw: sound strategy, small slips.

1 Major logical gap: unsupported conclusion.

0 Wrong skill: incompatible or premature.

COMMON TRAPS

Boundary leak → 2 Unproved tightness → 1

TAKEAWAY

Structured

A prototype specifies the skill, expected reasoning pattern, and concrete failure modes.

Reusable

The same rubric evaluates comparable subgoals across different problems.

Actionable

Reward reflects whether an intermediate behavior is complete—not merely whether it resembles a reference.

Reusable criteria reduce reward drift across instances.

Reference Quality Matters for Instance-Level Rubrics

RaR builds a rubric from each problem’s reference answer.
SCRIBE reuses skill rubrics without requiring golden reference answers.

Method	Reference source	MATH500	AIME25	BFCL
RaR	GPT-5.1 labeled	90.7	43.3	46.2
SCRIBE	Not required			

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Same Input, Same Rubric: Does the Score Stay Stable?

Repeat one fixed evaluation 10 times with independent judge sampling.

One Fixed Input

Problem: Find the maximum of $x(1 - x)$, $0 \leq x \leq 1$.

Trace: “ $x(1 - x) \leq 1/4$, so the maximum is $1/4$.”

Missing step: show that $x = 1/2$ attains the bound.

SCRIBE criterion: an upper bound alone does not establish a maximum.

Same problem + trace + judge
Same rubric, when applicable

10 evaluations $\rightarrow r_1, r_2, \dots, r_{10}$

Measure exact score agreement
and reward standard deviation.

Method	Agreement $\uparrow$ (%)	Reward Std. $\downarrow$
No-rubric PRM	71.8	0.63
RaR	82.4	0.41
SCRIBE	93.7	0.18

Illustrative input; table reports the paper's aggregate experiment, not this example's scores.

Different Problems, Same Behavior: Is Scoring Consistent?

Match subgoals by reasoning pattern or failure type, then compare their scores.

Problem A: Maximize $x(1 - x)$

" $x(1 - x) \leq 1/4$, so the maximum is $1/4$."

No check that equality is attainable.

Problem B: Maximize xy , Given $x + y = 10$

" $xy \leq 25$ by AM–GM, so the maximum is 25."

No check that equality is attainable.

Shared behavior: unproved tightness

Compare only the relevant subgoal spans.

One Matched Behavior Group

Score both spans under each method.

SCRIBE: one shared skill rubric.

RaR: the relevant dimension from each problem's rubric.

Compare score equality and $|r_A - r_B|$.

50 groups; 2–11 subgoals per group.

Method	Agree $\uparrow$	Gap $\downarrow$
Direct PRM	58.6%	0.82
RaR	69.4%	0.57
SCRIBE	86.8%	0.24

Reusable Rubrics Make Reward Judgments More Consistent

Compare no-rubric Process Reward Modeling, instance-level rubrics (RaR), and reusable skill rubrics (SCRIBE).

Same input, repeated 10 times

Fix the problem, trajectory, rubric, and judge.

Method	Agree ↑ (%)	Std. ↓
No-rubric PRM	71.8	0.63
RaR	82.4	0.41
SCRIBE	93.7	0.18

Similar behavior, different problems

50 matched behavior groups; 2–11 subgoals each.

Method	Agree ↑ (%)	Gap ↓
Direct PRM	58.6	0.82
RaR	69.4	0.57
SCRIBE	86.8	0.24

One shared rubric gives similar reasoning behaviors a more consistent reward standard.

Improved Reasoning on Math Tasks

Model	Method	MATH500	AIME25
Qwen3-4B Instruct-2507	Base	89.1	43.3
	Process Reward Modeling	92.3	51.7
	SCRIBE (Ours)		
LLaMA 3.2-3B-Instruct	Base	40.8	1.7
	Process Reward Modeling	48.3	6.7
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	SCRIBE (Ours)	63.4	15.8

BFCL V4: Evaluating Tool-Calling Behavior

WHAT IT TESTS

Can a model select the correct function, construct valid arguments, and coordinate multiple calls?

OUTPUT

One or more executable function calls—not a free-form final answer.

Berkeley Function Calling Leaderboard · V4

ILLUSTRATIVE TASK FORMAT

**User request + available function schemas
→ choose tool(s) → fill arguments → order calls**

Web Search
Retrieve information through search tools

Memory
Store and retrieve user-specific information

Single
Execute one live or non-live function call

Multi
Coordinate multiple calls for one request

Why SCRIBE: reliable process rewards must recognize complete intermediate tool-use behavior.

Improved Reasoning on General Tool-Use Tasks

Method	MATH500	AIME25	BFCL Overall	Search	Memory	Single Live	Single Non-Live	Multi
Qwen3-4B-Instruct-2507								
Base	89.1	43.3	33.0	5.0	12.7	86.7	74.5	19.4
Process Reward Modeling	92.3	51.7	44.6	10.0	15.2	88.5	77.6	28.6
SCRIBE (Ours)	95.8	63.3						
LLaMA 3.2-3B-Instruct								
Base	40.8	1.7	21.5	0.5	5.2	82.0	58.3	3.9
Process Reward Modeling	48.3	6.7	24.8	5.0	8.3	83.8	59.6	8.2
SCRIBE (Ours)	63.4	15.8						

Improved Reasoning on General Tool-Use Tasks

Method	MATH500	AIME25	BFCL Overall	Search	Memory	Single Live	Single Non-Live	Multi
Qwen3-4B-Instruct-2507								
Base	89.1	43.3	33.0	5.0	12.7	86.7	74.5	19.4
Process Reward Modeling	92.3	51.7	44.6	10.0	15.2	88.5	77.6	28.6
SCRIBE (Ours)	95.8	63.3	51.3	12.5	15.8	90.0	80.5	33.3
LLaMA 3.2-3B-Instruct								
Base	40.8	1.7	21.5	0.5	5.2	82.0	58.3	3.9
Process Reward Modeling	48.3	6.7	24.8	5.0	8.3	83.8	59.6	8.2
SCRIBE (Ours)	63.4	15.8	30.8	8.5	11.2	85.5	62.0	13.5

Takeaway: Reuse the Standard, Not the Reference Answer

Reward what a reasoning step accomplishes with a shared, reusable skill rubric.



[Project & Slides](#)

01

A Stable Standard Across Problems

Skill prototypes make reasoning quality explicit, without a golden answer for each instance.

02

Less Reward Drift

Cross-instance agreement: 69.4% with RaR → 86.8% with SCRIBE.

03

A Better Signal for Learning

Math and tool-use gains across both evaluated model families.

Build reusable criteria for intermediate behavior—not a new grading standard for every answer.